



Method of correlated objects

September 24, 2026

How does the concept of correlated objects work in detail? (Part of Tutorial 198).

The article has 4 parts:

- Introduction
- Basic Idea
- Proof of Concept
- Appendix: AI behind

Introduction

The concept of correlated objects links the local detection of an object with contextual knowledge about other objects in the same image. Thus, an object is evaluated not only based on its own appearance but also on whether its surroundings, location, size, and scene context are semantically plausible.[\[link.springer\]](#)[\[arxiv\]](#)

At its core, the described concept originates from probabilistic context classification or relational image analysis. Here, an image is not treated as a collection of independent objects; instead, the class assignment of a segment or object is subsequently reinforced or weakened based on its

neighbors and their probable classes.[kreissl]

Example of semantic segmentation

In an aerial image, an elongated gray segment might be ambiguous in isolation: it could be a road, a railway line, or a paved path.



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- If the system consistently detects parallel lines, gravel texture, and an elongated vehicle object nearby, the probability of it being a “railway track” increases.
- Conversely, if it detects traffic lanes, parked cars, and building entrances, “road” becomes more likely.
- If it detects adjacent bodies of water and bank vegetation, “waterside path” may become more plausible.

The neighboring elements do not “prove” the class, but they alter the probability distribution. This is the crucial difference compared to purely local object detection.



surroundings, location, size, and scene context

Historical Roots

The key conceptual sources lie in several closely related fields of research:

- **Pattern recognition and statistical decision theory:** Since the 1950s and 1960s, classification has been formulated as a probability problem: given observed features (x), the goal is to find the most probable class (c), for instance by determining ($P(c \mid x)$).
- **Markov Random Fields (MRF):** In the 1970s and 1980s, MRFs became a central model for image segmentation and scene analysis. Their key assumption is that the state of a pixel or image segment depends primarily on its local neighborhood. This gives rise to the concept that neighboring regions are “correlated” with one another.
- **Bayesian networks and probabilistic graphical models:** These allow for the joint representation of object classes, spatial relations, and prior knowledge. For example, “sky” appearing above a “building” is plausible, a “wheel” near a “car” is plausible, and a “rail” is more likely to be found next to a “locomotive” than next to a “cloud.”
- **Structural or relational pattern recognition:** Instead of simply comparing individual feature vectors, the scene is represented as a graph: nodes represent objects or segments, while edges represent

relationships such as adjacency, distance, contact, overlap, or “above.” Classification is then performed for the entire structure rather than for each object in isolation. [kreissl](#)

How it works in practice

A classic method operates in four steps:

1. The image is segmented—that is, broken down into contiguous regions or object candidates.
2. For each segment, a local classifier provides probabilities, such as: “gray elongated region = 55% road, 35% railway track, 10% river.”
3. A neighborhood graph is constructed from the segments.
4. An optimization or iteration process updates the class probabilities so that neighboring classes are mutually consistent.

Simplified, the rule mentioned is (Nachbarn=neighbor):

$$P(c_i | x_i, \text{Nachbarn}) \propto P(x_i | c_i) \cdot P(c_i | c_{\text{Nachbarn}})$$

Here, (x_i) denotes the visible features of an object, such as color, texture, and shape. The second term expresses contextual knowledge: how well does the object’s hypothesized class fit with the classes of neighboring objects?

Technical terms

Depending on the specific method, you may find this principle referred to by several names:

Term	Meaning
Contextual classification	Class decision incorporating the surrounding environment
Context-based object recognition	Utilization of other objects and spatial arrangement
Markov Random Field / MRF	Local neighborhood model based on probabilities

Term	Meaning
Conditional Random Field / CRF	Conditional graphical model, frequently used for segmentation and labeling
Relational or structural pattern recognition	Objects and their relationships are classified as a structure or graph
Region Adjacency Graph (RAG)	Graph composed of image regions and their adjacencies
Spatial-context model	Model utilizing knowledge regarding location, distance, and neighborhood

The approach you described—where probabilities are iteratively increased or decreased based on compatibility with neighboring objects—corresponds most directly to graph-based context classification using a region adjacency graph. [kreissl](#)

Relationship to modern AI (see Appendix A)

Modern deep learning models have not replaced this principle but have often implicitly adopted it:

- Convolutional Neural Networks (CNN) capture local image context through their expanding receptive fields and feature maps.
- Segmentation networks such as U-Net or DeepLab combine local details with large-scale scene context screening.
- CRFs have been—and sometimes still are—used as a post-processing step.

Basic idea explained



The cars make the fence more likely.

A detector initially provides a class, a bounding box, and a local confidence score for each detected region:

Candidate	Local Class	Local Confidence
Region A	Fence	0.58
Region B	Car	0.94
Region C	Road	0.89

Without context, “Fence” with a score of 0.58 would be merely an uncertain detection. The system then evaluates the data:

- Does the “fence” class frequently appear alongside cars and roads in such scenes?
- Is the car located in an expected position relative to the fence?
- Do the fence and car have plausible sizes and spacing?
- Does the overall pattern of objects fit the presumed scene—such as a parking lot, industrial site, or border facility?

If these relationships occurred frequently in the training data, the system can

assign a higher confidence level to the fence hypothesis. The correlation is therefore not merely a matter of visual similarity, but a learned statistical dependency between object classes, positions, and scenes. Co-occurrence specifically refers to joint appearance; spatial context encompasses distance, direction, orientation, overlap, adjacency, and relative position.

Correlations Used

In practice, several types are combined:

Context Type	Question	Example
Semantic Coexistence	Which classes frequently appear together?	Car and road; locomotive and track
Spatial Relation	Where is object B located relative to object A?	A signal is positioned near or next to a track
Distance/Adjacency	Are the objects sufficiently close or connected?	Carriages are positioned along the same track line
Relative Size	Is the size ratio appropriate?	A car is significantly smaller than a parking area
Orientation	Do the directions align?	Vehicles and traffic lanes run in similar directions
Scene Context	Does the object fit the overall scene?	Ship in water/harbor rather than a residential area
Hierarchical Context	Does the "part-to-whole" relationship fit?	Wheels, windows, roof, and track surroundings support "train"

For instance, a car is not credible as a car simply because it looks rectangular and metallic. The hypothesis becomes more plausible if the object is located on a road, is of an appropriate size, and appears near other vehicles.

Conversely, a context identified with high confidence can make weaker object candidates visible again. Studies on context detection therefore explicitly distinguish between semantic coexistence, spatial relationships, and size context.

Mathematical Model

Assume there are n object candidates. For candidate i , the system seeks a class label y_i —such as “fence,” “car,” or “road.” It then maximizes not merely the individual probability $P(y_i|x_i)$ (where x_i represents local image features) but the joint evaluation of all labels:

$$P(y_1, \dots, y_n \mid x_1, \dots, x_n, R)$$

R represents the spatial relationships between the candidates, such as distance, direction, and overlap. A common formulation as an optimization function is:

$$\operatorname{argmax}_y \left[\sum_i U_i(y_i, x_i) + \lambda \sum_{(i,j) \in E} V_{ij}(y_i, y_j, r_{ij}) + \mu G(y, S) \right]$$

$$\operatorname{yargmax}_i \sum U_i(y_i, x_i) + \lambda \sum_{(i,j) \in E} V_{ij}(y_i, y_j, r_{ij}) + \mu G(y, S)$$

Here, the following apply:

- U_i : local evidence from the detector for candidate i (e.g., “this segment looks 58% like a fence”).
- V_{ij} : pairwise context term for two objects i and j ; it evaluates whether the classes and spatial relationship are compatible.
- r_{ij} : measured relationship (e.g., “car 3 m in front of fence,” “object to the left of the road,” or “parallel to the track axis”).
- E : set of object pairs considered—often limited to spatial neighbors to maintain computational efficiency.
- G : global scene term that evaluates the set of objects against a scene type such as “train station,” “harbor,” “parking lot,” or “residential area.”
- λ and μ : weights determining the relative importance of local detection,

pairwise context, and the overall scene.

Conditional Random Fields (CRFs) are a classic approach for precisely this task. They model a large set of interdependent labels and are used in semantic segmentation to explicitly account for correlations between pixels or regions. The CoLA (Co-occurrence, Location, and Appearance) method, for instance, combines local image features, co-occurrence, and relative positions within a CRF to select labels that are consistent with both the segment content and each other.

Step by step

- Generating object candidates

A detector such as YOLO, Faster R-CNN, or a segmentation model provides bounding boxes or regions, class probabilities, and features. A candidate can have multiple plausible labels, not just its top-1 result.

- Calculating relational features

For each relevant pair, the system calculates, for example:

$r_{ij} = (\Delta x, \Delta y, \text{distance, overlap, size ratio, angle difference})$

In GeoAI, georeferenced distance, elevation, proximity to roads/tracks, land-cover class, and temporal change are also included.

- Learning correlations or defining rules
- A table is generated from annotated images or geospatial data, for example:

$P(\text{Track} \mid \text{Locomotive})$ high

$P(\text{Road} \mid \text{Car})$ high

$P(\text{Catenary mast near track} \mid \text{Electrified railway})$ high

In doing so, a statistical model learns not only that two classes co-occur but also typical relationships: "Object B is located within specific positional and distance ranges relative to A."

- Updating scores

A new score is generated based on local evidence and context.

Visually:

$$\text{Score_new}(\text{fence}) = \text{Score_local}(\text{fence}) + \lambda \cdot \text{Plausibility}(\text{fence}, \text{car}, r)$$

If the car is actually in a position expected for the domain, the score increases. If it is far away, on the wrong side, or in an unsuitable environment, the bonus remains small or becomes negative.

- Reaching a joint decision

Because A influences B and B, in turn, influences A, a single correction step is often insufficient. Classical models use message passing or belief propagation for this purpose; neighboring nodes iteratively pass on evidence until the probabilities stabilize. An example of a hierarchical CRF process involves representing regions as nodes, weighting edges based on their similarity, and subsequently performing belief propagation.

- Threshold and plausibility check

Only then is a decision made regarding whether a detection is output, suppressed, reclassified, or flagged for human review. This is important: a context bonus should not automatically turn a weak detection into apparent certainty.

Example: Railway Infrastructure

A domain-specific graph would be particularly effective for aerial or drone imagery:

Object	Expected correlations
Track	Parallel to another track; connected to switches; near ballast bed
Catenary mast	At regular intervals and near a track axis
Platform	Parallel to the track; near station roof or access point
Train	Elongated object group, aligned with the track axis

Object	Expected correlations
Signal	Close to the track; usually in a typical orientation

Suppose the model identifies a series of slender vertical objects, each with a confidence score of only 0.45, as “catenary masts.” If it simultaneously detects a track with high confidence and the candidates are situated parallel to it at roughly regular intervals, this combination can reinforce the mast hypotheses. Conversely, if the same posts are found in a row of trees far from the track, such a boost would be inappropriate.

Limitations and robust implementation

Context serves as an additional source of evidence, not a substitute for visual proof. Three risks are particularly relevant:

- **Spurious correlations:** A model might learn arbitrary training patterns—such as consistently associating cars with specific camera angles or regions, or interpreting a car near a fence as a **spurious correlation** indicating a parking lot.
- **Domain shift:** Patterns considered normal in training images do not necessarily apply to other countries, seasons, sensors, or historical aerial imagery.
- **Error propagation:** An incorrect detection with a high confidence score can erroneously “confirm” neighboring detections.

The method becomes robust if it tracks local and contextual probabilities separately, utilizes relationships only within meaningful distances, limits the influence of the context term, and flags unusual combinations for review rather than forcibly reclassifying them. A recent overview of context-based object detection highlights the importance of pairwise spatial relationships, co-occurrence statistics, and their combination with graphical models such as CRFs.[\[link.springer\]](#)

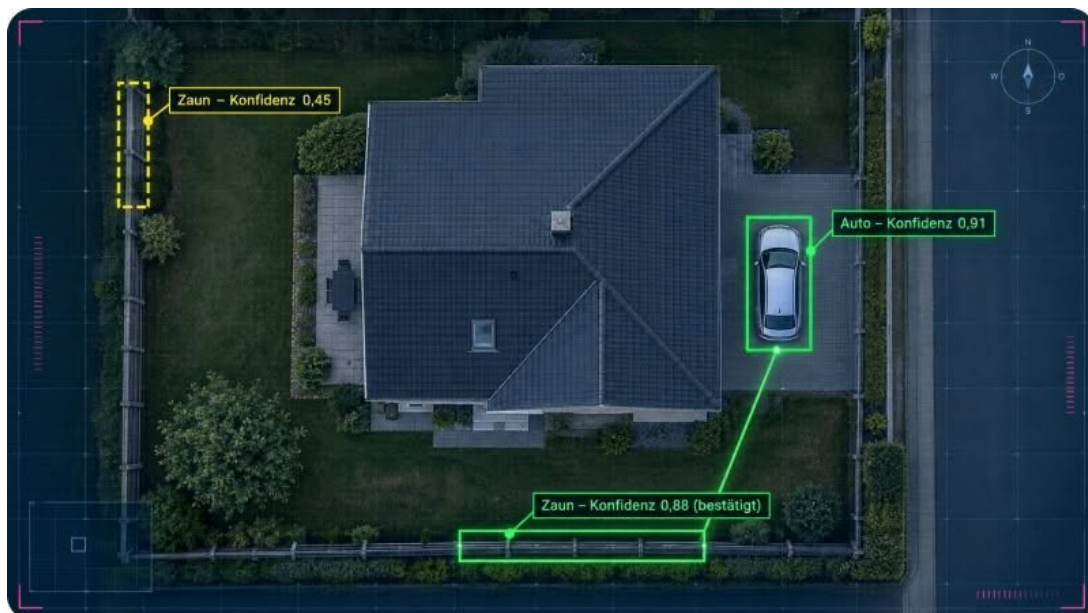
Semantic segmentation using probabilistic models assigns a class probability to every pixel in an image and subsequently determines the most plausible overall labeling for all pixels collectively. Unlike isolated pixel classification, the model therefore takes into account image content, spatial

proximity, color and texture similarity, and object boundaries. Conditional Random Fields (CRFs) and Markov Random Fields (MRFs) are the classic models used for this purpose.

Proof of concept

Which applies: “if car, then fence” or “if fence, then car”?

There is no blanket rule of “car $\Rightarrow$ fence” or “fence $\Rightarrow$ car.” In correlation-based image analysis, both conditional probabilities can be modeled—but they are typically different and do not constitute causal claims.



Evidence and Hypothesis

There is no blanket rule of “Car $\Rightarrow$ Fence” or “Fence $\Rightarrow$ Car.” In a correlation-based image analysis, **both conditional probabilities** can be modeled—though they are typically different and do not constitute causal claims.
[plato.stanford](https://plato.stanford.edu)

The Right Direction

When examining a fence candidate with low confidence—given that a car has already been detected with relatively high confidence—the relevant question is:

$$[P(\text{Fence}|\text{Car}, r)]$$

that is: *How likely is a fence, given that a car has been observed in spatial relation (r)?*

This corresponds to the case:

Car as evidence $\Rightarrow$ Re-evaluate fence score.

Conversely, if the car detection is uncertain but a fence has been detected with high confidence, you examine:

$$P(\text{Auto}|\text{Zaun}, r)$$

$$[P(\text{Car} \mid \text{Fence}, r)]$$

that is:

Fence as evidence $\Rightarrow$ Re-evaluate car score.

Both directions represent valid inference steps. Which one you use depends on which object serves as the evidence and which serves as the hypothesis to be confirmed.

The Probabilities Are Asymmetric

The two statements are not equivalent:

$$P(\text{Fence}|\text{Car}) \neq P(\text{Car}|\text{Fence}).$$

This is because cars appear in many situations where no fence is visible: streets, parking garages, public squares, highways, city centers, or rural roads. Consequently, a car is often only a weak indicator of a fence.

A fence, on the other hand, can more frequently serve as an indicator of a nearby car—for instance, at parking lots, industrial sites, depots, border installations, or private properties. However, this also depends on the training area, viewing angle, map section, and object definition. For instance, a long pasture fence is not a strong indicator of the presence of cars.

Evidence	Hypothesis Tested	Typical Statement
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Evidence	Hypothesis Tested	Typical Statement
Car detected	Fence uncertain	“Does a car at this position increase the probability of a fence?”
Fence detected	Car uncertain	“Does a fence at this position increase the probability of a car?”
Car and fence detected	Overall scene	“Is the combination—including geometry—plausible?”

Crucial: The Spatial Relationship

In a useful model, a simple class-based relationship is insufficient. Instead of just

$$P(\text{Zaun}|\text{Auto})$$

$$[P(\text{Fence}|\text{Car})]$$

one actually needs:

$$P(\text{Zaun}|\text{Auto}, r, s).$$

$$[P(\text{Fence}|\text{Car}, r, s).]$$

Here:

- (r): denotes the specific relationship—distance, direction, side, contact, relative size, orientation.
- (s): denotes the scene context—for example, a parking lot, a street, a train station, agricultural land, or an industrial area.

In practical terms, this could mean:

$$[P(\text{Fence} | \text{Car}, \text{Car 2–8 m in front of the candidate, Parking space})]$$

is high, whereas

[$P(\text{fence} \mid \text{car}, \text{car 150 m away, open highway})$]

remains close to the general baseline value.

Thus, the context model should not learn the illogical rule “Every car implies a fence,” but rather a conditional, geometrically constrained relationship.

Bayesian Perspective

It is often easier to know or learn the probability of seeing a car **given that a fence is actually present**.

However, to actually confirm the presence of a fence given an observed car, the reverse conditional probability is required:

[$P(\text{fence} \mid \text{car}, r)$.]

Bayes’ theorem provides the conversion:

$$P(\text{Zaun} \mid \text{Auto}, r) = \frac{P(\text{Auto} \mid \text{Zaun}, r) \cdot P(\text{Zaun} \mid r)}{P(\text{Auto} \mid r)}.$$

$\frac{P(\text{car} \mid \text{fence}, r) \cdot P(\text{fence} \mid r)}{P(\text{car} \mid r)}$.

The prior $P(\text{Zaun} \mid r)$ ($P(\text{fence} \mid r)$) is particularly significant here: even if cars are frequently found near fences, a fence might be rare overall within a specific image domain. In that case, a single car will only marginally increase the fence score. Bayes’ theorem links the “direct” conditional probability ($P(H \mid E)$) with the inverse probability ($P(E \mid H)$) and prior knowledge about the hypothesis. [plato.stanford](https://plato.stanford.edu/)

During inference, evidence flows in **both directions** along the edge:

- a high-confidence car detection can reinforce a fence candidate;
- a high-confidence fence detection can reinforce a car candidate;
- two low-confidence candidates can provide moderate mutual support;
 - two conflicting candidates or an implausible geometry weaken the combination.

This is not a logical “if-then” relationship, but rather a weighted contribution to consistency. The value is optimized in conjunction with local CNN/detector scores, rather than forcing the derivation of one class from the other.

Important: Correlation does not imply causation.

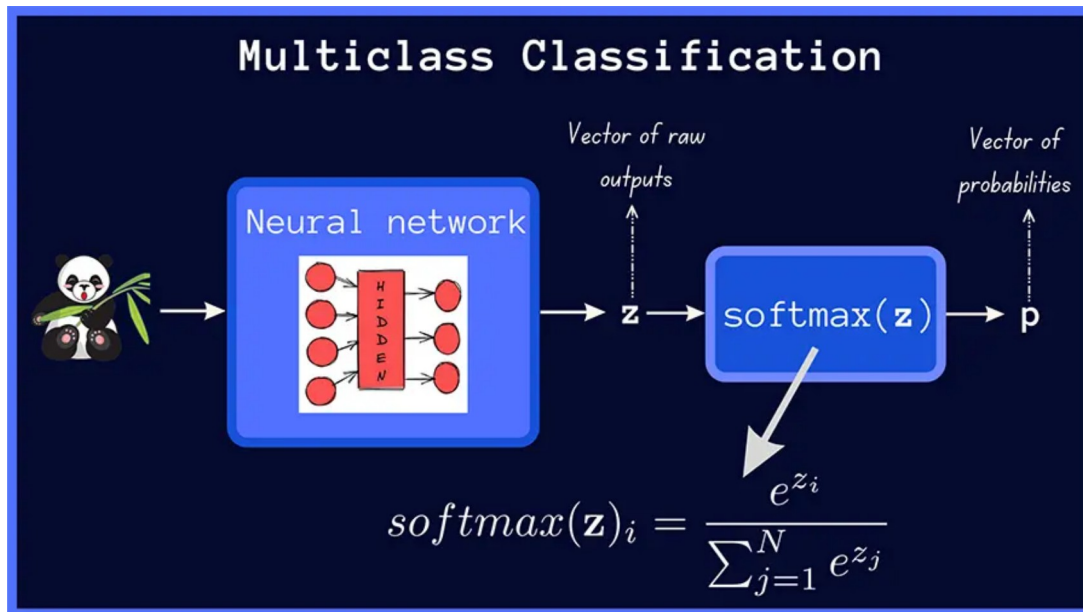
The fact that “cars frequently appear near fences” does not mean that cars create fences or that fences cause cars. Both may appear together due to a third, not directly observed cause—such as a parking lot, industrial area, secured site, or private property acting as a source of spurious correlation. Probabilistic dependence alone cannot determine the direction of causality; shared third-party causes can also explain the observed correlation.
[plato.stanford]

The practical rule is therefore:

Use the object detected with higher confidence and appropriate spatial positioning as evidence to re-weight the conditional probability of an uncertain object candidate—in the direction relevant to the specific question at hand.

Appendix: Corr. Objects Learning in a Neural Network (under construction! in 9 steps)

In the neural network, these “probabilities” do not arise from hard-coded rules—such as “car means fence”—but rather from trained weights. The network first calculates uncalibrated values (*logits*) for each pixel, segment, or object candidate; only a softmax or sigmoid function converts these into values that can be interpreted as class probabilities. [openaccess.thecvf](https://openaccess.thecvf.com/)



Softmax function in multiclass classification

1. From Image to Features

An input image is processed in multiple stages by a CNN, Vision Transformer, or hybrid network, resulting in the creation of feature maps.

For a pixel, an image patch, or an object region, a feature vector might look like this:

$$\mathbf{f}_i = (1, 8, -0, 4, 2, 1, \dots, 0, 7).$$

The individual dimensions do not represent immediately readable properties such as "gray," "long," or "next to the track." Instead, they encode learned visual patterns—such as edges, textures, geometries, material appearances, and increasingly complex object parts.

In segmentation, every pixel or spatial location has such a vector. In detection, region proposal networks or query-based detectors additionally generate object candidates, including their location and size.

2. Logits: The Raw Values

The final classification layer calculates a raw value for each class per candidate:

$$z_{i,c} = \mathbf{w}_c^T \mathbf{f}_i + b_c.$$

Here:

- (f_i) : learned features for pixel, region, or object candidate (i)
- (w_c) : weight vector for class (c)
- (b_c) : class bias
- $(z_{\{i,c\}})$: logit for class (c)

For three classes, a network might output, for example:

Class	Logit
Fence	2.1
Car	0.8
Vegetation	-0.5

A logit is not yet a probability. It can be negative, does not have to sum to 1, and is primarily a relative evidence score.

3. Softmax generates class distributions

For mutually exclusive classes—such as “road,” “track,” “vegetation,” or “building”—Softmax is used:

$$P(y_{i,c} = 1 \mid I, \theta) = \sigma(z_{i,c}) = \frac{1}{1 + \exp(-z_{i,c})} \cdot$$

$$\frac{\exp(z_{\{i,c\}})}{\sum_{k=1}^K \exp(z_{\{i,k\}})} .$$

The Softmax output satisfies:

$$0 \leq P(y_i = c \mid I, \theta) \leq 1$$

$$0 \leq P(y_i = c \mid I, \theta) \leq 1$$

[$0 \leq P(y_i = c \mid I, \theta) \leq 1$] and

$$\sum_{c=1}^K P(y_i = c \mid I, \theta) = 1.$$

$$\sum_{c=1}^K P(y_i = c \mid I, \theta) = 1.$$

$$[\sum_{c=1}^K P(y_i = c \mid I, \theta) = 1.]$$

Based on the example logits, the result is approximately:

Class	Softmax probability
Fence	0.75
Car	0.20
Vegetation	0.05

The network would select “Fence” because it has the highest probability.

However, in object detection, classes can overlap: a pixel might simultaneously fit the categories “vehicle” and “military vehicle,” or an image region might possess multiple attributes. In this case, a model often employs an independent sigmoid function for each class:

$$\tilde{z}_i = f_i + \sum_{j \neq i} \alpha_{ij} W_v f_j.$$

$$\frac{1}{1 + \exp(-z_{i,c})}.$$

Unlike Softmax, the sigmoid function does not force the class values to sum to 1.

4. How the network learns

During training, each image is associated with reference data (*ground truth*):

- For segmentation: a class mask per pixel.
- For detection: object class plus bounding box, mask, or keypoints.

- For context learning: additionally, relationships, scene labels, or—indirectly—numerous examples featuring typical spatial arrangements.

For a pixel correctly annotated as “fence,” for instance, the network should increase $P(\text{fence})$. The standard loss function is cross-entropy:

$$L_i = -\log P(y_i = c^* \mid I, \theta),$$

$$-\log P(y_i = c^* \mid I, \theta),$$

where (c^*) is the correct class.

If the fence is correctly assigned a probability of 0.95, the loss is low:

$$[-\log(0.95) \approx 0.051.]$$

If the probability for the fence is only 0.10, the loss is high:

$$[-\log(0.10) \approx 2.303.]$$

Using backpropagation, the training process calculates which weights contributed to this error. An optimizer such as SGD or Adam adjusts them in small steps. When repeated across many annotated examples, this yields a parameter vector (θ) that approximates the conditional distribution

$$P(y \mid I, \theta)$$

$$[P(y \mid I, \theta).]$$

In this context, combining Softmax with cross-entropy corresponds to directly learning a discriminative class probability for each pixel or region.

[openaccess.thecvf](https://openaccess.thecvf.com/)

5. Where Object Correlation Comes In

There are three common levels—ranging from implicit to explicit.

Approach	How context is incorporated	Result
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Approach	How context is incorporated	Result
Large receptive field	Deep CNN layers “see” a larger image area	The network can learn: typical street or parking lot structure makes a car more plausible
Attention / Transformer	Image regions or object queries weight each other	An object candidate can incorporate information from other regions
Relation module / Graph network / CRF	Relationships betw	

...candidates are explicitly modeled | Local scores are updated using class, positional, and scene relations |

In a detection network with a relation module, a set of feature vectors $((f_1, \dots, f_n))$ is derived from multiple object candidates. For candidate (i), information from the other candidates is aggregated:

$$\tilde{f}_i = f_i + \sum_{j \neq i} \alpha_{ij} W_v f_j.$$

$f_i + \sum_{j \neq i} \alpha_{ij} W_v f_j$

- $(W_v f_j)$ is the transformed information from object (j).
- (α_{ij}) is a learned attention value: How relevant is object (j) to the assessment of (i)?
- (f_i) is retained so that the context supplements local visual evidence rather than replacing it.

Relation Networks for object detection utilize precisely this concept: object relationships are used to re-evaluate detections; examples cited include co-occurrence and spatial relationships. [openaccess.thecvf](https://openaccess.thecvf.com/)

6. How α_{ij} is calculated

An attention module generates three representations from each object candidate:

Darstellungen:

$$q_i = W_q f_i, \quad k_j = W_k f_j, \quad v_j = W_v f_j.$$

Additionally, the geometry between the two boxes is captured:

$$g_{ij} = \left(\frac{x_j - x_i}{w_i}, \frac{y_j - y_i}{h_i}, \log \frac{w_j}{w_i}, \log \frac{h_j}{h_i} \right).$$

$$g_{ij} = (w_i x_j - x_i, h_i y_j - y_i, \log w_i w_j, \log h_i h_j).$$

$$[g_{ij}] = \left(\frac{x_j - x_i}{w_i}, \frac{y_j - y_i}{h_i}, \log \frac{w_j}{w_i}, \log \frac{h_j}{h_i} \right).$$

The attention value can then be calculated as follows:

$$\alpha_{ij} = \text{softmax}_j \left(\frac{q_i^\top k_j}{\sqrt{d}} + h(g_{ij}) \right)$$

$$\text{softmax}_j \left(\frac{q_i^\top k_j}{\sqrt{d}} + h(g_{ij}) \right).$$

$h(g_{ij})$ is a small neural network that learns which geometries are relevant.

This allows the model to learn that:

- An object directly next to a candidate is more relevant than one far away.
- A small object on a road can represent a vehicle context.
- Elongated, parallel elements can reinforce each other in the context of a track.

- An object in an unsuitable location makes a minimal or even indirectly negative contribution.

Subsequently, a classification layer calculates new logits and softmax/sigmoid probabilities based on $(\tilde{f}_i)$. Importantly, the network does not need to explicitly store a rule like “Car $\Rightarrow$ Fence.” It can learn from examples that certain visual and geometric configurations tend to increase or decrease the fence logit on average.

7. Example involving a fence and a car

Consider a fence candidate (i) with a local feature vector (f_{fence}) . Without context, it receives:

$$z_{\text{Zaun}} = 0,4.$$

$$z_{\text{Zaun}} = 0,4.$$

A reliably detected car candidate (j) is located in a position typical of the training data. The relation unit yields approximately:

$$\alpha_{\text{Zaun,Auto}} = 0,32.$$

$$[\alpha_{\text{fence}, \text{car}} = 0.32.]$$

After context aggregation, the fence logit may increase:

$$z'_{\text{Zaun}} = z_{\text{Zaun}} + \Delta z_{\text{Kontext}} = 0,4 + 0,9 = 1,3.$$

1.3.]

The final probability depends on all competing classes. With Softmax, the value for the fence can thus rise from 0.42 to 0.71, for example.

These figures are merely illustrative. In a real-world model, (α) , the weights, and the score contribution are not selected manually but are learned from data.

8. Probabilistic models following the network

A classic alternative or complement involves the following steps:

1. The segmentation network generates class probabilities for each pixel.
2. These are converted into unary energies:

$$\psi_u(y_i = c) = -\log P_{\text{Netz}}(y_i = c \mid I).$$

$$\psi_u(y_i=c)=-\log P_{\text{Netz}}(y_i=c|I).$$

$$[\psi_u(y_i=c)=-\log P_{\{\text{Net}\}}(y_i=c \mid I).]$$

3. A CRF connects pixels or regions and evaluates label combinations based on spatial proximity, color, texture, and—where applicable—object relationships.
4. Iterative inference produces a new, spatially more consistent probability map.

Here, the network provides the local evidence, while the CRF handles explicit neighborhood and boundary modeling. In CRF-RNN approaches, the inference steps can even be formulated as a recurrent network layer and trained end-to-end alongside the CNN. [openaccess.thecvf](https://openaccess.thecvf.com/)

9. “Probability” does not automatically imply certainty

Technically, a Softmax value of 0.95 means that the model assigns a very high score to the class relative to its other classes. It does not automatically mean that the prediction is correct in 95 out of 100 real-world cases.

Neural networks can be overconfident, particularly with:

- unfamiliar sensors, camera heights, or resolutions
- different countries, landscapes, or seasons
- rare objects and unusual viewing angles
- data outside the training distribution. For robust applications, **calibration** is therefore verified—for instance, using a separate validation/test dataset and methods such as temperature scaling. Additionally, local evidence, contextual contribution, and final confidence should be stored separately—a practice particularly valuable in fields like GEOINT, infrastructure inspection, or safety-critical decision-making.

Appendix German Translation

Lernen im neuronalen Netz

Die „Wahrscheinlichkeiten“ entstehen im neuronalen Netz nicht durch fest programmierte Regeln wie „Auto bedeutet Zaun“, sondern aus trainierten Gewichten. Das Netz berechnet zunächst für jedes Pixel, jedes Segment oder jeden Objektkandidaten ungeeichte Werte (*Logits*); erst eine Softmax- oder Sigmoid-Funktion macht daraus Werte, die als Klassenwahrscheinlichkeiten interpretiert werden können.

Für drei Klassen könnte ein Netz beispielsweise ausgeben:

Klasse	Logit
Zaun	2,1
Auto	0,8
Vegetation	-0,5

Ein Logit ist noch keine Wahrscheinlichkeit. Er darf negativ sein, muss sich nicht auf 1 summieren und ist vor allem ein relativer Evidenzscore.

Aus den Beispiel-Logits ergibt sich ungefähr:

Klasse	Softmax-Wahrscheinlichkeit
Zaun	0,75
Auto	0,20
Vegetation	0,05

Das Netz würde „Zaun“ wählen, weil dessen Wahrscheinlichkeit am größten ist.

Bei Objektdetektion können Klassen einander aber überlappen: Ein Pixel kann gleichzeitig zu „Fahrzeug“ und „militärisches Fahrzeug“ passen, oder ein

Bildausschnitt kann mehrere Attribute haben. Dann verwendet ein Modell oft pro Klasse eine unabhängige Sigmoid-Funktion:

$$P(y_{i,c} = 1 \mid I, \theta) = \sigma(z_{i,c}) = \frac{1}{1 + \exp(-z_{i,c})}.$$

$$P(y_{i,c}=1 \mid I, \theta) = \sigma(z_{i,c}) = \frac{1}{1 + \exp(-z_{i,c})}.$$

Anders als Softmax zwingt Sigmoid die Klassenwerte nicht dazu, zusammen 1 zu ergeben.

4. Wie das Netz lernt

Beim Training besitzt jedes Bild Referenzdaten (*Ground Truth*):

- Für Segmentierung: eine Klassenmaske pro Pixel.
- Für Detektion: Objektklasse plus Bounding Box, Maske oder Keypoints.
- Für Kontextlernen: zusätzlich Beziehungen, Szene-Labels oder indirekt viele Beispiele, in denen typische räumliche Anordnungen auftreten.

Für ein korrekt als Zaun annotiertes Pixel soll das Netz beispielsweise $P(\text{Zaun})$ erhöhen. Die Standard-Loss-Funktion ist Cross-Entropy:

$$L_i = -\log P(y_i = c^* \mid I, \theta),$$

$$L_i = -\log P(y_i = c^* \mid I, \theta),$$

wobei c^* die richtige Klasse ist.

Ist der Zaun korrekt mit 0,95 bewertet, ist der Verlust klein:

$$-\log(0,95)$$

$$-\log(0,95) \approx 0,051.$$

Liegt die Zaunwahrscheinlichkeit nur bei 0,10, ist der Verlust groß:

$$-\log(0,10) \approx 2,303.$$

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Mittels Backpropagation berechnet das Trainingsverfahren, welche Gewichte zu diesem Fehler beigetragen haben. Ein Optimierer wie SGD oder Adam passt sie in kleinen Schritten an. Wiederholt über viele annotierte Beispiele entsteht ein Parametervektor θ , der die bedingte Verteilung

$$P(y \mid I, \theta)$$

$$P(y||\theta)$$

approximiert. Softmax plus Cross-Entropy entspricht dabei dem direkten Lernen einer diskriminativen Klassenwahrscheinlichkeit pro Pixel oder Region.
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5. Wo Objektkorrelation hineinkommt

Es gibt drei verbreitete Stufen — von implizit bis explizit.

Ansatz	Wie der Kontext eingeht	Ergebnis
Großes Rezeptivfeld	Tiefe CNN-Schichten „sehen“ einen größeren Bildbereich	Das Netz kann lernen: typische Straßen- oder Parkplatzstruktur macht ein Auto plausibler
Attention / Transformer	Bildregionen bzw. Objekt-Queries gewichten einander	Ein Objektkandidat kann Information anderer Regionen aufnehmen
Relationsmodul / Graphnetz / CRF	Beziehungen zwischen Kandidaten werden ausdrücklich modelliert	Lokale Scores werden mithilfe von Klassen-, Lage- und Szenenrelationen aktualisiert

In einem Detektionsnetz mit Relationsmodul wird aus mehreren Objektkandidaten ein Satz von Merkmalsvektoren $f_1, \dots, f_n$. Für Kandidat i wird Information der anderen Kandidaten aggregiert:

$$\tilde{f}_i = f_i + \sum_{j \neq i} \alpha_{ij} W_v f_j.$$

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- $W_v f_j$ $W_v f_j$ ist die transformierte Information von Objekt j .
- α_{ij} α_{ij} ist ein gelernter Aufmerksamkeitswert: Wie relevant ist Objekt j für die Bewertung von i ?
- f_i f_i bleibt erhalten, damit der Kontext lokale visuelle Evidenz ergänzt und nicht ersetzt.

Relation Networks für Objektdetektion nutzen genau diesen Gedanken: Objektbeziehungen werden für die Neubewertung von Detektionen verwendet; als Beispiele nennen sie gemeinsame Vorkommen und räumliche Beziehungen. [openaccess.thecvf](https://openaccess.thecvf.com/)

6. Wie α_{ij} α_{ij} berechnet wird

Ein Attention-Modul bildet aus jedem Objektkandidaten drei Darstellungen:

$$q_i = W_q f_i, \quad k_j = W_k f_j, \quad v_j = W_v f_j.$$

$$q_i = W_q f_i, k_j = W_k f_j, v_j = W_v f_j.$$

Zusätzlich wird die Geometrie zwischen beiden Boxen erfasst:

$$g_{ij} = \left(\frac{x_j - x_i}{w_i}, \frac{y_j - y_i}{h_i}, \log \frac{w_j}{w_i}, \log \frac{h_j}{h_i} \right).$$

$$g_{ij} = (w_i x_j - x_i, h_i y_j - y_i, \log w_i w_j, \log h_i h_j).$$

Der Aufmerksamkeitswert kann dann so berechnet werden:

$$\alpha_{ij} = \text{softmax}_j \left(\frac{q_i^\top k_j}{\sqrt{d}} + h(g_{ij}) \right).$$

$$\alpha_{ij} = \text{softmax}_j (d q_i^\top k_j + h(g_{ij})).$$

$h(g_{ij})$ $h(g_{ij})$ ist ein kleines neuronales Netz, das lernt, welche Geometrien relevant sind. Dadurch kann das Modell lernen:

- Ein Objekt direkt neben einem Kandidaten ist relevanter als eines weit entfernt.
- Ein kleines Objekt auf einer Straße kann ein Fahrzeugkontext sein.
- Längliche, parallele Elemente können sich im Bahnkontext gegenseitig stützen.
- Ein Objekt in unpassender Lage liefert einen geringen oder sogar indirekt negativen Beitrag.

Danach berechnet eine Klassifikationsschicht aus $\tilde{f}_i$ neue Logits und Softmax-/Sigmoid-Wahrscheinlichkeiten. Wichtig: Das Netz muss nicht explizit als Regel gespeichert haben „Auto $\Rightarrow$ Zaun“. Es kann aus Beispielen lernen, dass bestimmte visuelle und geometrische Konstellationen den Zaun-Logit im Mittel erhöhen oder senken.

7. Beispiel mit Zaun und Auto

Nehmen wir einen Zaun-Kandidaten i mit lokalem Featurevektor f_{Zaun} . Ohne Kontext erhält er:

$$z_{\text{Zaun}} = 0,4.$$

$$z_{\text{Zaun}} = 0,4.$$

Ein sicher erkannter Auto-Kandidat j liegt in einer aus den Trainingsdaten typischen Position. Die Relationseinheit liefert etwa:

$$\alpha_{\text{Zaun,Auto}} = 0,32.$$

$$\alpha_{\text{Zaun,Auto}} = 0,32.$$

Nach Kontextaggregation kann sich der Zaun-Logit erhöhen:

$$z'_{\text{Zaun}} = z_{\text{Zaun}} + \Delta z_{\text{Kontext}} = 0,4 + 0,9 = 1,3.$$

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Die endgültige Wahrscheinlichkeit hängt von allen konkurrierenden Klassen ab. Mit Softmax kann der Zaun dadurch zum Beispiel von 0,42 auf 0,71 steigen.

Diese Zahlen sind nur eine Illustration. In einem realen Modell werden α , die Gewichte und der Scorebeitrag nicht von Hand gewählt, sondern aus Daten gelernt.

8. Probabilistische Modelle nach dem Netz

Eine klassische Alternative oder Ergänzung ist:

1. Das Segmentierungsnetz erzeugt pro Pixel Klassenwahrscheinlichkeiten.
2. Diese werden in Unary-Energien überführt:

$$\psi_u(y_i = c) = -\log P_{\text{Netz}}(y_i = c \mid I).$$

$$\psi_u(y_i=c)=-\log P_{\text{Netz}}(y_i=c \mid I).$$

3. Ein CRF verbindet Pixel oder Regionen und bewertet Labelkombinationen anhand von Bildnähe, Farbe, Textur und gegebenenfalls Objektrelationen.
4. Iterative Inferenz erzeugt eine neue, räumlich konsistentere Wahrscheinlichkeitskarte.

Das Netz liefert hier die lokale Evidenz; das CRF übernimmt die explizite Nachbarschafts- und Grenzmodellierung. In CRF-RNN-Ansätzen lassen sich die Inferenzschritte sogar als wiederkehrende Netzschicht formulieren und zusammen mit dem CNN Ende-zu-Ende trainieren.[openaccess.thecvf](https://openaccess.thecvf.com/)

9. „Wahrscheinlichkeit“ ist nicht automatisch Sicherheit

Ein Softmaxwert von 0,95 bedeutet technisch: Das Modell gibt der Klasse relativ zu seinen anderen Klassen eine sehr hohe Bewertung. Es bedeutet nicht automatisch, dass die Aussage in 95 von 100 realen Fällen korrekt sein muss.

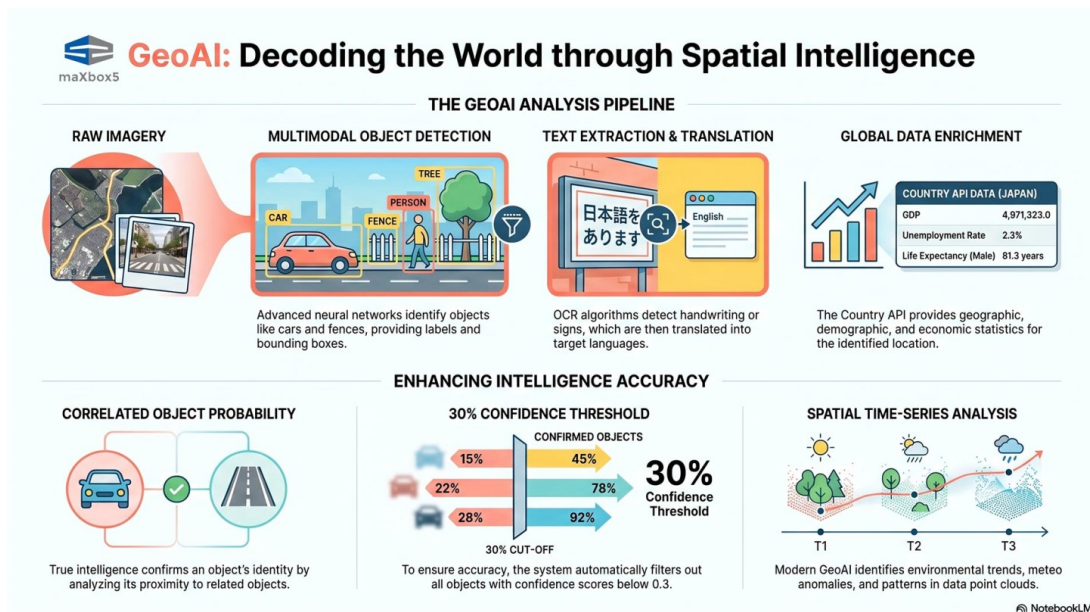
Neural Netze können überkonfident sein, besonders bei:

- unbekannten Sensoren, Kamerahöhen oder Auflösungen
- anderen Ländern, Landschaften oder Jahreszeiten
- seltenen Objekten und ungewöhnlichen Blickwinkeln
- Daten außerhalb der Trainingsverteilung.

Für belastbare Anwendungen prüft man deshalb **Kalibrierung**, beispielsweise mit einem getrennten Validierungs-/Testdatensatz und Verfahren wie Temperature Scaling. Zusätzlich sollten lokale Evidenz, Kontextbeitrag und finale Konfidenz getrennt gespeichert werden — besonders sinnvoll bei GEOINT, Infrastrukturinspektion oder sicherheitsrelevanten Entscheidungen.

Links & Literatur

[1] GEO Intelligence Infograph: Prompting, Max Kleiner, NotebookML



GEO Intelligence Infograph: Prompting, Max Kleiner, NotebookML

[2] Szenenanalyse, Max

Kleiner, [GEOINT_Screenshot_1-7-2026_135415_www.softwareschule.ch.jpg](https://www.softwareschule.ch/jpg)

[3] Codeblock, Max Kleiner: maXbox5/

EKON30/1473_Geo_Intelligence_API3_52.txt at main · maxkleiner/maXbox5

[4] Erweiterte Geointelligenz: Integration von Karten, APIs und KI für verbesserte Geodatenanalyse: <https://de.slideshare.net/slideshow/advanced-geo-intelligence-integrating-maps-apis-and-ai-for-enhanced-geospatial-analysis/288228122>

[5] EKON 30 Session – GEO Intelligence APIs: <https://maxkleiner1.medium.com/ekon-30-30cd211550de>

[6] Animationsvideo: <https://www.youtube.com/watch?v=E97OROtEkGg>

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